United States Patent	12411399
Kind Code	B2
Date of Patent	September 09, 2025
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Projection system and projector

Abstract

A projection system includes a first and second optical system arranged from a reduction side toward an enlargement side. The second optical system includes an optical element having a concave reflection surface and a first lens having negative power, the optical element and first lens arranged from reduction side toward enlargement side. The projection system satisfies the following expressions:

TR≤0.3 (1)

35≤(OAL/imy)×(LL/imy)×TR×(1/NA)≤60 (2)

OAL represents an axial inter-surface spacing from an image formation device to the reflection surface, imy represents a first distance from an optical axis to the largest image height at the image formation device, LL represents the largest radius of the first lens, TR represents a throw ratio, and NA represents the numerical aperture of the image formation device.

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Appl. No.: **18/156451**

Filed: **January 19, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230236491 A1	Jul. 27, 2023

Foreign Application Priority Data

JP	2022-006175	Jan. 19, 2022

Publication Classification

Int. Cl.: **G03B21/28** (20060101); **G02B13/16** (20060101)

U.S. Cl.:

CPC **G03B21/28** (20130101); **G02B13/16** (20130101);

Field of Classification Search

CPC: G02B (13/18); G02B (13/16); G02B (17/08); G02B (17/0844); G02B (17/0812); G02B (17/0852); G02B (17/0848); G02B (17/0828); G03B (21/10); G03B (21/28)

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-006175, filed Jan. 19, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a projection system and a projector.

2. Related Art

(3) JP-A-2020-34690 describes a projector in which a projection system enlarges a projection image displayed at an image display device and projects the enlarged projection image onto a screen. The projection system includes a first refractive optical system, a reflective optical system, and a second refractive optical system sequentially arranged from the reduction side toward the enlargement side. The first refractive optical system includes a plurality of refractive lenses. The reflective optical system includes a concave mirror and reflects beams from the first refractive optical system toward the side facing the image display device in directions that intersect with the

optical axis of the first refractive optical system. The second refractive optical system is formed of a single refractive lens. The refractive lens is an enlargement-side lens located at a position closest to the enlargement side in the projection system. Beams from the concave mirror enter the enlargement-side lens in directions that intersect with the optical axis of the enlargement-side lens.
(4) Out of the examples of the projection system disclosed in JP-A-2020-34690, the projection system having the shortest projection distance has a projection distance of 257.6 mm. The enlargement-side lens of the thus configured projection system has an effective radius of 79.7 mm. The thus configured projection system further has a throw ratio of 0.154.
(5) A projector including a projection system having a smaller throw ratio has a shorter projection distance over which the projector projects an enlarged image having a predetermined size. A projection system incorporated in a projector used indoors or at similar locations therefore needs to have a short focal length that provides a throw ratio smaller than or equal to 0.3.
(6) A projection system having a shorter focal length tends to produce larger amounts of aberrations at the enlargement side. It is therefore necessary to increase the effective radius of the enlargement-side lens, through which the beams from the concave mirror obliquely pass, to allow the enlargement-side lens to correct the beams on an image height basis. When the size of the enlargement-side lens is increased to provide a sufficient effective radius, however, the amount of protrusion by which the enlargement-side lens protrudes radially from the first optical axis of the first refractive optical system increases, resulting in an increase in the diameter of the entire projection system. The size of the projector that incorporates the projection system is therefore not reduced.

SUMMARY

(7) To solve the problem described above, a projection system according to an aspect of the present disclosure is a projection system for enlarging a projection image formed by an image formation device disposed in a reduction-side conjugate plane and projecting the enlarged image in an enlargement-side conjugate plane. The projection system includes a first optical system and a second optical system sequentially arranged from the reduction side toward the enlargement side. The first optical system includes a diaphragm. The second optical system includes an optical element having a concave reflection surface and a first lens having negative power, the optical element and the first lens sequentially arranged from the reduction side toward the enlargement side. An intermediate image conjugate with the reduction-side conjugate plane and the enlargement-side conjugate plane is formed between the first optical system and the second optical system. A portion at the reduction side of the first optical system forms a telecentric portion. The projection system satisfies all Conditional Expressions (1) and (2) below,

$TR \leq 0.3 \qquad (1)$

$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60 \qquad (2)$

where OAL represents an axial inter-surface spacing from the image formation device to the reflection surface, imy represents a first distance from an optical axis to a largest image height at the image formation device, LL represents a largest radius of the first lens, TR represents a throw ratio that is a quotient of division of a projection distance by a second distance from the optical axis to a largest image height of the enlarged image, and NA represents a numerical aperture of the image formation device.
(8) A projector according to another aspect of the present disclosure includes the projection system described above and the image formation device that forms a projection image in the reduction-side conjugate plane of the projection system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** shows a schematic configuration of a projector including a projection system according to an embodiment of the present disclosure.
(2) FIG. **2** is a beam diagram showing beams passing through the projection system according to Example 1.
(3) FIG. **3** shows lateral aberrations produced by the projection system according to Example 1 set at a standard distance.
(4) FIG. **4** shows spherical aberration, astigmatism, and distortion produced by the projection system according to Example 1 set at the standard distance.
(5) FIG. **5** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 1 set at a short distance.
(6) FIG. **6** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 1 set at a long distance.
(7) FIG. **7** is a beam diagram showing beams passing through the projection system according to Example 2.
(8) FIG. **8** shows the lateral aberrations produced by the projection system according to Example 2 set at the standard distance.
(9) FIG. **9** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 2 set at the standard distance.
(10) FIG. **10** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 2 set at the short distance.
(11) FIG. **11** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 2 set at the long distance.
(12) FIG. **12** is a beam diagram showing beams passing through the projection system according to Example 3.
(13) FIG. **13** shows the lateral aberrations produced by the projection system according to Example 3 set at the standard distance.
(14) FIG. **14** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 3 set at the standard distance.
(15) FIG. **15** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 3 set at the short distance.
(16) FIG. **16** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 3 set at the long distance.
(17) FIG. **17** is a beam diagram showing beams passing through the projection system according to Example 4.
(18) FIG. **18** shows the lateral aberrations produced by the projection system according to Example 4 set at the standard distance.
(19) FIG. **19** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 4 set at the standard distance.
(20) FIG. **20** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 4 set at the short distance.
(21) FIG. **21** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 4 set at the long distance.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(22) An optical system and a projector according to an embodiment of the present disclosure will be described below with reference to the drawings.

Projector

(23) FIG. **1** shows a schematic configuration of a projector including a projection system **3** according to the embodiment of the present disclosure. A projector **1** includes an image formation

unit **2**, which generates a projection image to be projected onto a screen S, the projection system **3**, which enlarges the projection image and projects the enlarged image onto the screen S, and a controller **4**, which controls the operation of the image formation unit **2**, as shown in FIG. **1**.

Image Formation Unit and Controller

(24) The image formation unit **2** includes a light source **10**, a first optical integration lens **11**, a second optical integration lens **12**, a polarization converter **13**, and a superimposing lens **14**. The light source **10** is formed, for example, of an ultrahigh-pressure mercury lamp or a solid-state light source. The first optical integration lens **11** and the second optical integration lens **12** each include a plurality of lens elements arranged in an array. The first optical integration lens **11** divides a luminous flux from the light source **10** into a plurality of luminous fluxes. The lens elements of the first optical integration lens **11** focus the luminous flux from the light source **10** in the vicinity of the lens elements of the second optical integration lens **12**.

(25) The polarization converter **13** converts the light from the second optical integration lens **12** into predetermined linearly polarized light. The superimposing lens **14** superimposes images of the lens elements of the first optical integration lens **11** on one another in a display area of each of liquid crystal panels **18**R, **18**G, and **18**B, which will be described later, via the second optical integration lens **12**.

(26) The image formation unit **2** further includes a first dichroic mirror **15**, a reflection mirror **16**, a field lens **17**R, and the liquid crystal panel **18**R. The first dichroic mirror **15** reflects R light, which is part of the beams incident via the superimposing lens **14**, and transmits G light and B light, which are part of the beams incident via the superimposing lens **14**. The R light reflected off the first dichroic mirror **15** travels via the reflection mirror **16** and the field lens **17**R and is incident on the liquid crystal panel **18**R. The liquid crystal panel **18**R is an image formation element. The liquid crystal panel **18**R modulates the R light in accordance with an image signal to form a red projection image.

(27) The image formation unit **2** further includes a second dichroic mirror **21**, a field lens **17**G, and the liquid crystal panel **18**G. The second dichroic mirror **21** reflects the G light, which is part of the beams via the first dichroic mirror **15**, and transmits the B light, which is part of the beams via the first dichroic mirror **15**. The G light reflected off the second dichroic mirror **21** passes through the field lens **17**G and is incident on the liquid crystal panel **18**G. The liquid crystal panel **18**G is an image formation element. The liquid crystal panel **18**G modulates the G light in accordance with an image signal to form a green projection image.

(28) The image formation unit **2** further includes a relay lens **22**, a reflection mirror **23**, a relay lens **24**, a reflection mirror **25**, a field lens **17**B, the liquid crystal panel **18**B, and a cross dichroic prism **19**. The B light having passed through the second dichroic mirror **21** travels via the relay lens **22**, the reflection mirror **23**, the relay lens **24**, the reflection mirror **25**, and the field lens **17**B and is incident on the liquid crystal panel **18**B. The liquid crystal panel **18**B is an image formation element. The liquid crystal panel **18**B modulates the B light in accordance with an image signal to form a blue projection image.

(29) The liquid crystal panels **18**R, **18**G, and **18**B surround the cross dichroic prism **19** so as to face three sides of the cross dichroic prism **19**. The cross dichroic prism **19** is a prism for light combination and produces a projection image that is the combination of the light modulated by the liquid crystal panel **18**R, the light modulated by the liquid crystal panel **18**G, and the light modulated by the liquid crystal panel **18**B.

(30) The projection system **3** enlarges the combined projection image from the cross dichroic prism **19** and projects the enlarged projection image onto the screen S.

(31) The control unit **4** includes an image processor **6**, to which an external image signal, such as a video signal, is inputted, and a display driver **7**, which drives the liquid crystal panels **18**R, **18**G, and **18**B based on image signals outputted from the image processor **6**.

(32) The image processor **6** converts the image signal inputted from an external apparatus into

image signals each containing grayscales and other factors of a color corresponding to the image signal. The display driver **7** operates the liquid crystal panels **18**R, **18**G, and **18**B based on the color projection image signals outputted from the image processor **6**. The image processor **6** thus causes the liquid crystal panels **18**R, **18**G, and **18**B to display projection images corresponding to the image signals.

Projection System

(33) The projection system **3** will next be described. The screen S is disposed in the enlargement-side conjugate plane of the projection system **3**, as shown in FIG. **1**. The liquid crystal panels **18**R, **18**G, and **18**B are disposed in the reduction-side conjugate plane of the projection system **3**.

(34) Examples 1 to 4 will be described below as examples of the configuration of the projection system **3** incorporated in the projector **1**.

Example 1

(35) FIG. **2** is a beam diagram showing beams passing through a projection system **3**A according to Example 1. In the beam diagrams for the projection systems **3** according to Examples 1 to 4, the liquid crystal panels **18**R, **18**G, and **18**B are referred to as a liquid crystal panel **18**. The projection system **3**A according to the present example is formed of a first optical system **31** and a second optical system **32** sequentially arranged from the reduction side toward the enlargement side, as shown in FIG. **2**. The second optical system **32** is disposed on an optical axis N of the first optical system **31**.

(36) In the following description, three axes perpendicular to one another are called axes X, Y, and Z for convenience. The axis Z coincides with the optical axis N of the first optical system **31**. The direction along the optical axis N is an axis-Z direction. The axis-Z direction toward the side where the first optical system **31** is located is called a first direction Z**1**, and the axis-Z direction toward the side where the second optical system **32** is located is called a second direction Z**2**. The axis Y extends along the screen S. The upward-downward direction is an axis-Y direction, with one side of the axis-Y direction called an upper side Y**1** and the other side of the axis-Y direction called a lower side Y**2**. The axis X extends in the width direction of the screen.

(37) The first optical system **31** is a refractive optical system. The first optical system **31** is formed of 17 lenses L**1** to L**17**. The lenses L**1** to L**17** are arranged in the presented order from the reduction side toward the enlargement side. A diaphragm **51** is disposed between the lens L**7** and the lens L**8**.

(38) The lens L**6** has aspherical shapes at opposite sides. The lens L**9** has aspherical shapes at opposite sides. The lens L**16** (third lens) has aspherical shapes at opposite sides. The lens L**17** (second lens) has aspherical shapes at opposite sides. The lens L**2** and the lens L**3** are bonded to each other into a cemented doublet L**21**. The lens L**4** and the lens L**5** are bonded to each other into a cemented doublet L**22**. The lens L**11** and the lens L**12** are bonded to each other into a cemented doublet L**23**. The lens L**14** and the lens L**15** are bonded to each other into a cemented doublet L**24**.

(39) The second optical system **32** includes an optical element **33** and a first lens **34**. The optical element **33** and the first lens **34** are arranged in the presented order from the reduction side toward the enlargement side. The optical element **33** has a reflection surface **40**, which faces the reduction side. The reflection surface **40** has a concave shape recessed in the second direction Z**2**. The reflection surface **40** has an aspherical shape. The reflection surface **40** is located at the lower side Y**2** of the optical axis N, as shown in FIG. **2**. The reflection surface **40** is formed by providing the outer surface, in the first direction Z**1**, of the optical element **33** with a reflection coating layer (reflection layer). The reflection surface **40** reflects light at the surface, facing in the direction Z**1**, of the optical element **33**.

(40) The first lens **34** is located at a position shifted from the optical element **33** in the first direction Z**1** and disposed at the upper side Y**1** of the optical axis N. The first lens **34** has negative power. The first lens **34** has a convex enlargement-side surface and a concave reduction-side surface. The first lens **34** has aspherical shapes at opposite sides.

(41) The liquid crystal panel **18** of the image formation unit **2** is disposed in the reduction-side

conjugate plane of the projection system **3**A. The screen S is disposed in the enlargement-side conjugate plane of the projection system **3**A.

(42) The liquid crystal panel **18** forms a projection image in an image formation plane perpendicular to the optical axis N of the first optical system **31**. The liquid crystal panel **18** is disposed in a position offset from the optical axis N of the first optical system **31** toward the upper side Y**1**. The projection image is therefore formed in a position offset from the optical axis N toward the upper side Y**1**.

(43) The beams from the liquid crystal panel **18** pass through the first optical system **31** and the second optical system **32** in the presented order. Between the first optical system **31** and the second optical system **32**, the beams pass through the lower side Y**2** of the optical axis N. The beams are therefore directed through the second optical system **32** toward the reflection surface **40**. The beams having reached the reflection surface **40** are deflected back in the first direction Z**1** towards the upper side Y**1**. The beams deflected back by the reflection surface **40** cross the optical axis N toward the upper side Y**1** and travels toward the first lens **34**. The beams passing through the first lens **34** are widened by the first lens **34** and reach the screen S.

(44) The lens L**17** of the first optical system **31** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. An intermediate image **30** is formed between the lens L**17** and the reflection surface **40**.

(45) In the projection system **3**A, the portion at the reduction side of the first optical system **31** is a telecentric portion. The term “telecentric” means that the central beam of each luminous flux traveling between the first optical system **31** and the liquid crystal panel **18** disposed in the reduction-side conjugate plane is parallel or substantially parallel to the optical axis of the projection system.

(46) The projection system **3**A has a changeable projection distance. When the projection distance is changed, eight lenses of the first optical system **31**, the lenses L**10** to L**17**, are moved along the optical axis N for focusing. In the focusing, the lenses L**10**, L**11**, and L**12** are moved as a unit. In the focusing, the lenses L**13**, L**14**, and L**15** are moved also as a unit.

(47) Data on the projection system **3**A are listed below,

(48) TABLE-US-00001 NA 0.3125 imy 11.7 mm scy 1463 mm PD 283.1 mm M 125 TR 0.194 OAL 203 mm LL 47.8 mm

where NA represents the numerical aperture of the liquid crystal panel **18**, imy represents a first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, scy represents a second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, PD represents a projection distance that is the distance from the first lens **34** to the screen S, M represents a projection magnification that is the quotient of division of the second distance by the first distance, TR represents a throw ratio that is the quotient of division of the projection distance by the second distance, OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, and LL represents the largest radius of the first lens **34**.

(49) Data on the lenses of the projection system **3**A are listed below. The surfaces of the lenses are numbered sequentially from the reduction side toward the enlargement side. Reference characters are given to the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen. Data labeled with a surface number that does not correspond to any of the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen is dummy data. Reference character R represents the radius of curvature. Reference character D represents the axial inter-surface spacing. Reference character C represents the aperture radius, and twice the aperture radius is the diameter of the lens surface. Reference characters R, D, and C are each expressed in millimeters.

(50) TABLE-US-00002 Reference Surface Refraction/ character number Shape R D Glass material Reflection C 18 0 Spherical Infinity 9.5000 Refraction 0.0000 19 1 Spherical Infinity 25.9100

SBSL7_OHARA Refraction 13.0371 2 Spherical Infinity 0.0000 Refraction 15.4259 L1 3 Spherical 23.6460 8.4106 SFPL51_OHARA Refraction 16.3497 4 Spherical −163.6273 0.1000 Refraction 16.1238 L2 5 Spherical 24.1479 5.7206 SFSL5_OHARA Refraction 14.0000 L3 6 Spherical 113.0998 1.0000 STIH6_OHARA Refraction 13.1404 7 Spherical 34.7820 0.1000 Refraction 12.2395 L4 8 Spherical 19.4148 7.7264 SBSL7_OHARA Refraction 11.6173 L5 9 Spherical −25.1932 0.9500 TAFD25_HOYA Refraction 10.9949 10 Spherical 25.2418 0.2000 Refraction 10.0083 L6 11 Aspherical 16.8372 4.4033 LBAL35_OHARA Refraction 10.1020 12 Aspherical 40.5023 1.0000 Refraction 9.1836 L7 13 Spherical 19.4289 2.3917 SFSL5_OHARA Refraction 9.1301 14 Spherical 32.7745 4.0203 Refraction 8.8594 51 15 Spherical Infinity 0.1043 Refraction 8.3951 L8 16 Spherical 101.2882 3.7504 STIH53_OHARA Refraction 8.5137 17 Spherical −22.7544 0.1000 Refraction 8.6718 L9 18 Aspherical −21.2771 6.0000 LLAM60_OHARA Refraction 8.6423 19 Aspherical 103.2802 Variable Refraction 9.5014 spacing 1 L10 20 Spherical 22.9132 2.0346 STIM22_OHARA Refraction 10.9384 21 Spherical 26.8895 14.2089 Refraction 10.9386 L11 22 Spherical 57.1566 5.3002 STIM2_OHARA Refraction 16.5000 L12 23 Spherical −129.3659 1.0000 STIH6_OHARA Refraction 16.7439 24 Spherical 677.5558 Variable Refraction 17.0582 spacing 2 L13 25 Spherical 47.4289 9.6511 STIM22_OHARA Refraction 18.3999 26 Spherical −46.5818 0.1000 Refraction 18.3651 L14 27 Spherical −67.5767 8.1053 STIL25_OHARA Refraction 17.8500 L15 28 Spherical −21.6948 1.0000 STIH6_OHARA Refraction 17.7177 29 Spherical 170.7327 Variable Refraction 18.7445 spacing 3 L16 30 Aspherical −24.7550 3.0000 ‘Z-E48R’ Refraction 18.8159 31 Aspherical 101.4895 Variable Refraction 21.3950 spacing 4 L17 32 Aspherical 257.2804 8.0000 ‘Z-E48R’ Refraction 25.6568 33 Aspherical 63.8637 Variable Refraction 28.1976 spacing 5 40 34 Aspherical −28.5800 −50.8918 Reflection 34 35 Aspherical 59.7480 −7.0000 ‘Z-E48R’ Refraction 37.2465 36 Aspherical 67.7207 Variable Refraction 47.7801 spacing 6 S 37 Spherical Infinity 0.0000 Refraction 1985.1150

(51) The projection system **3**A according to the present example has a changeable projection distance selected from a standard distance, a short distance shorter than the standard distance, and a long distance longer than the standard distance. When the projection distance is changed, eight lenses of the first optical system **31**, the lenses L**10** to L**17**, are moved along the optical axis N for focusing. When the focusing is performed so as to change the projection distance from the short distance to the long distance, the lenses L**10**, L**11**, and L**12** move along the optical axis N toward the enlargement side. In the same focusing operation, the lenses L**13**, L**14**, and L**15** move along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**16** moves along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**17** moves along the optical axis N toward the reduction side.

(52) The table below shows the variable spacings 1, 2, 3, 4, 5, and 6 at the projection distances where the focusing is performed. The variable spacing 1 is the axial inter-surface spacing between the lens L**9** and the lens L**10**. The variable spacing 2 is the axial inter-surface spacing between the lens L**12** and the lens L**13**. The variable spacing 3 is the axial inter-surface spacing between the lens L**15** and the lens L**16**. The variable spacing 4 is the axial inter-surface spacing between the lens L**16** and the lens L**17**. The variable spacing 5 is the axial inter-surface spacing between the lens L**17** and the reflection surface **40**. The variable spacing 6 is the projection distance.

(53) TABLE-US-00003 Standard Short Long distance distance distance Variable spacing 1 1.1069 0.8768 1.2979 Variable spacing 2 0.8042 0.1000 1.4634 Variable spacing 3 6.2530 6.4325 6.0906 Variable spacing 4 20.9176 21.6588 20.1541 Variable spacing 5 39.8875 39.8713 39.9334 Variable spacing 6 −283.0000 −217.0000 −401.0000

(54) The aspherical coefficients are listed below.

(55) TABLE-US-00004 Surface number S11 S12 S18 S19 Radius of 16.8372 40.5023 −21.2771 103.2802 curvature (R) Conic constant (K) 8.76367E−01 1.40552E+01 −1 −90 Fourth-order −2.95336E−06 8.91728E−05 4.09548E−05 7.14087E−05 Sixth-order 5.97014E−08 2.72315E−07 −4.12331E−07 −3.41490E−07 Eighth-order 7.38171E−11

−7.34670E−10 1.32697E−09 1.10129E−09 Tenth-order 1.42398E−11 Surface number S30 S31 S32 S33 Radius of −24.7550 101.4895 257.2804 63.8637 curvature (R) Conic constant (K) 0 0 90 0.00000E+00 Fourth-order 9.42799E−05 1.68772E−05 −4.34174E−05 −6.79148E−05 Sixth-order −3.84902E−07 −1.78130E−07 1.23752E−07 1.63921E−07 Eighth-order 8.80106E−10 3.67946E−10 −2.74764E−10 −2.83439E−10 Tenth-order −8.42504E−13 −3.17825E−13 3.77647E−13 2.91903E−13 Twelfth-order −2.03100E−16 −1.17046E−16 Fourteenth-order 6.56482E−22 Surface number S34 S35 S36 Radius of −28.5800 59.7480 67.72067982 curvature (R) Conic constant (K) −1.00000E+00 2.18390E−01 −0.004229729 Fourth-order 3.62586E−06 2.99552E−05 1.47026E−05 Sixth-order −6.05961E−09 −9.57606E−08 −3.80264E−08 Eighth-order 7.45065E−12 1.64656E−10 4.62847E−11 Tenth-order −5.45419E−15 −1.61839E−13 −3.18524E−14 Twelfth-order 2.08318E−18 9.28635E−17 1.29283E−17 Fourteenth-order −3.40484E−22 −2.85578E−20 −2.89715E−21 Sixteenth-order 3.61610E−24 2.80551E−25

(56) The projection system **3**A according to the present example satisfies all Conditional Expressions (1) and (2) below,

$TR \leq 0.3$ (1)

$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60$ (2)

where OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, imy represents the first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, LL represents the largest radius of the first lens **34**, TR represents the throw ratio, which is the quotient of division of the projection distance by the second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, and NA represents the numerical aperture of the liquid crystal panel **18**.

(57) Furthermore, it is more preferable that the projection system **3**A satisfies all Conditional Expressions (1) and (2′) below.

$TR \leq 0.3$ (1)

$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 53$ (2′)

(58) In the present example, the values described above are listed below.

(59) TABLE-US-00005 OAL 203 mm imy 11.7 mm LL 47.8 mm TR 0.194 NA 0.3125

(60) TR=0.194 is provided from the table shown above, so that Conditional Expression (1) is satisfied. (OAL/imy)×(LL/imy)×TR×(1/NA)=44 is satisfied, so that Conditional Expression (2) is satisfied.

Effects and Advantages

(61) The projection system **3**A according to the present example enlarges a projection image formed by the liquid crystal panel **18** disposed in the reduction-side conjugate plane and projects the enlarged projection image in the enlargement-side conjugate plane. The projection system **3**A according to the present example includes the first optical system **31** and the second optical system **32** sequentially arranged from the reduction side toward the enlargement side. The first optical system **31** includes the diaphragm **51**. The second optical system **32** includes the optical element **33**, which has the concave reflection surface **40**, and the first lens **34**, which has negative power, sequentially arranged from the reduction side toward the enlargement side. The intermediate image **30** conjugate with the reduction-side conjugate plane and the enlargement-side conjugate plane is formed between the first optical system **31** and the second optical system **32**. The portion at the reduction side of the first optical system **31** form a telecentric portion.

(62) The projection system **3**A according to the present example satisfies all Conditional Expressions (1) and (2) below,

$TR \leq 0.3$ (1)

$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60$ (2)

where OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, imy represents the first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, LL represents the largest radius of the first lens **34**, TR represents the throw ratio, which is the quotient of division of the projection distance by the second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, and NA represents the numerical aperture of the liquid crystal panel **18**.

(63) The projection system **3**A according to the present example satisfies Conditional Expression (1). The projection system **3** therefore has a short focal length. A projection system having a shorter focal length tends to produce larger amounts of aberrations at the enlargement side. It is therefore necessary to increase the effective radius of the enlargement-side lens, through which the beams from the concave mirror obliquely pass, to allow the enlargement-side lens to correct the beams on an image height basis. When the size of the enlargement-side lens is increased to provide a sufficient effective radius, however, the amount of protrusion by which the enlargement-side lens protrudes radially from the first optical axis of the first refractive optical system increase, resulting in an increase in the diameter of the entire projection system.

(64) To solve the problem described above, the projection system **3**A according to the present example satisfies Conditional Expression (2). Suppression of the amount of protrusion by which the first lens **34** protrudes radially from the optical axis N can therefore suppress an increase in the diameter of the entire projection system, whereby the size of the projector that incorporates the projection system **3**A can be reduced. Furthermore, the effective diameter of the first lens **34** within which the beams can be corrected on an image height basis can be ensured, while the amount of protrusion by which the first lens **34** protrudes radially from the optical axis N is suppressed. That is, when (OAL/imy)×(LL/imy)×TR×(1/NA) in Conditional Expression (2) is smaller than the lower limit, the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40** and the lens diameter of the first lens **34** become too small relative to TR and 1/NA, so that it is difficult to correct the beams on an image height basis, and sufficient resolution of the projection system **3**A is unlikely to be provided. Even when a lens that can provide sufficient resolution can be designed, the lens has a problem of low mass producibility because the lens needs to be manufactured with high molding precision. When (OAL/imy)×(LL/imy)×TR×(1/NA) in Conditional Expression (2) is greater than the upper limit, the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40** and the lens diameter of the first lens **34** become excessively large. That is, the amount of protrusion by which the first lens **34** protrudes radially from the optical axis N increases, resulting in an increase in the diameter of the entire projection system. The size of the projector that incorporates the projection system therefore increases.

(65) Example 3 described in JP-A-2020-34690, which is a related-art literature, will now be examined as Comparable Example. The projection system according to Comparable Example includes a first refractive optical system, a reflective optical system, and a second refractive optical system sequentially arranged from the reduction side toward the enlargement side. The first refractive optical system includes a plurality of refractive lenses. The reflective optical system includes a concave mirror and reflects beams from the first refractive optical system toward the side facing the image display device in directions that intersect with the optical axis of the first refractive optical system. The second refractive optical system is formed of a single refractive lens. The refractive lens is an enlargement-side lens located at a position closest to the enlargement side in the projection system. Beams from the concave mirror enter the enlargement-side lens in directions that intersect with the optical axis of the enlargement-side lens. Data on Comparable Example are listed below.

(66) TABLE-US-00006 OAL 256 mm imy 13.2 mm LL 79.7 mm TR 0.154 NA 0.25

(67) In Comparable Example, TR=0.154. The projection system according to Comparable Example therefore satisfies Conditional Expression (1). In Comparable Example, however, (OAL/imy)×(LL/imy)×TR×(1/NA)=72. The projection system according to Comparable Example therefore does not satisfy Conditional Expression (2). Therefore, when the throw ratio is fixed in the present example and Comparative Example, the effective radius of the enlargement-side lens of

the projection system according to Comparative Example is greater than the effective radius of the first lens of the projection system **3**A according to the present example. That is, the entire projection system according to Comparative Example has a diameter greater than that of the entire projection system **3**A according to the present example.

(68) In the projection system **3**A according to the present example, the reflection surface **40** is provided with a reflection coating layer (reflection layer). In the configuration in which the reflection surface is provided inside the optical element **33**, the accuracy of the shape of the enlargement-side lens surface, at which the reflection surface is provided, depends on the accuracy of the shape of the optical element **33**. That is, to improve the accuracy of the shape of the enlargement-side lens surface, the accuracy of the shape of the reduction-side lens surface also needs to be improved. In contrast, since the reflection surface **40** of the projection system **3**A according to the present example is provided at the outer surface of the optical element **33**, only the accuracy of the shape of the outer surface of the optical element **33** needs to be improved. The accuracy of the shape of the reflection surface **40** in the present example is therefore readily improved as compared with that in the configuration in which the reflection surface is provided inside the optical element **33**.

(69) In the configuration in which the reflection surface is provided inside the optical element **33**, the optical element **33** is formed, and a reflection coating layer is then formed at the enlargement-side lens surface of the optical element **33** to form the reflection surface. In this process, a support film layer needs to be provided between the reflection coating layer and the enlargement-side lens surface. Although the thus provided support film layer causes the reflection coating layer to be unlikely to peel off the enlargement-side lens surface, the interposed support film layer tends to lower the optical performance of the reflection surface, so that the optical performance of the reflection surface tends to vary in the manufacturing process. In contrast, in the projection system **3**A according to the present example, the support film layer is provided on the side opposite from the reflection surface of the reflection coating layer, whereby the optical performance of the reflection surface **40** is unlikely to deteriorate. Stable optical performance of the reflection surface **40** is therefore likely to be achieved during the manufacture of the optical element **33**.

(70) In the projection system **3**A according to the present example, the lens L**17** (second lens) disposed at a position closest to the enlargement side in the first optical system **31** is formed as a lens separate from the first lens **34**. The lens L**17** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. That is, the lens L**17** disposed at a position closest to the enlargement side in the first optical system **31** is disposed inside the second optical system **32** in the direction of the optical axis N, so that the distance between the lens L**17** and the reflection surface **40** decreases. The axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40** can thus be shortened, whereby the size of the projection system **3**A can be reduced. Furthermore, the intermediate image **30** is formed between the lens L**17** of the first optical system **31** and the reflection surface **40** of the second optical system **32**. When the distance between the lens L**17** and the reflection surface **40** thus decreases, a variety of aberrations contained in the intermediate image **30** are readily corrected on an image height basis.

(71) The first optical system **31** includes the lens L**17** (second lens) and the lens L**16** (third lens), which is disposed adjacent to the lens L**17** and shifted therefrom toward the reduction side. The lenses L**16** and L**17** each have an aspherical shape. In the projection system **3**A according to the present example, focusing that causes the projection distance to be changed from the short distance to the long distance is performed by moving the lenses L**16** and L**17** toward the enlargement side in the direction of the optical axis N. The projection system **3**A, in which the lenses L**16** and L**17**, which correct a variety of aberrations on an image height basis, are moved in the direction of the optical axis N, therefore allows suppression of occurrence of the variety of aberrations during focusing. Furthermore, in a configuration in which a lens having no aspherical shape is moved in the direction of the optical axis N for focusing, an aspherical lens that corrects a variety of

aberrations needs to be separately prepared. In contrast, the present example, in which the lenses L**16** and L**17**, which move during focusing, each have an aspherical shape, allows reduction in the size of the entire projection system.
(72) The first optical system **31** further includes the cemented doublets L**23** and L**24** at the enlargement side of the diaphragm **51**. The chromatic aberrations can therefore be corrected well.
(73) FIG. **3** shows lateral aberrations produced by the projection system **3**A set at the standard distance. FIG. **4** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**A set at the standard distance. FIG. **5** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**A set at the short distance. FIG. **6** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**A set at the long distance. The projection system **3**A according to the present example produces an enlarged image having suppressed aberrations, as shown in FIGS. **3** to **6**.
Example 2
(74) FIG. **7** is a beam diagram showing beams passing through a projection system **3**B according to Example 2. The projection system **3**B according to the present example is formed of a first optical system **31** and a second optical system **32** sequentially arranged from the reduction side toward the enlargement side, as shown in FIG. **7**. The second optical system **32** is disposed on an optical axis N of the first optical system **31**.
(75) The first optical system **31** is a refractive optical system. The first optical system **31** is formed of sixteen lenses L**1** to L**16**. The lenses L**1** to L**16** are arranged in the presented order from the reduction side toward the enlargement side. A diaphragm **51** is disposed between the lens L**7** and the lens L**8**.
(76) The lens L**6** has aspherical shapes at opposite sides. The lens L**9** has aspherical shapes at opposite sides. The lens L**15** (third lens) has aspherical shapes at opposite sides. The lens L**16** (second lens) has aspherical shapes at opposite sides. The lens L**2** and the lens L**3** are bonded to each other into a cemented doublet L**21**. The lens L**4** and the lens L**5** are bonded to each other into a cemented doublet L**22**. The lens L**10** and the lens L**11** are bonded to each other into a cemented doublet L**23**. The lens L**13** and the lens L**14** are bonded to each other into a cemented doublet L**24**.
(77) The second optical system **32** includes an optical element **33** and a first lens **34**. The optical element **33** and the first lens **34** are arranged in the presented order from the reduction side toward the enlargement side. The optical element **33** has a reflection surface **40**, which faces the reduction side. The reflection surface **40** has a concave shape recessed in the second direction Z**2**. The reflection surface **40** has an aspherical shape. The reflection surface **40** is located at the lower side Y**2** of the optical axis N, as shown in FIG. **7**. The reflection surface **40** is formed by providing the outer surface, in the first direction Z**1**, of the optical element **33** with a reflection coating layer (reflection layer). The reflection surface **40** reflects light at the surface, facing in the direction Z**1**, of the optical element **33**.
(78) The first lens **34** is located at a position shifted from the optical element **33** in the first direction Z**1** and disposed at the upper side Y**1** of the optical axis N. The first lens **34** has negative power. The first lens **34** has a convex enlargement-side surface and a concave reduction-side surface. The first lens **34** has aspherical shapes at opposite sides.
(79) The liquid crystal panel **18** of the image formation unit **2** is disposed in the reduction-side conjugate plane of the projection system **3**B. The screen S is disposed in the enlargement-side conjugate plane of the projection system **3**B.
(80) The liquid crystal panel **18** forms a projection image in an image formation plane perpendicular to the optical axis N of the first optical system **31**. The liquid crystal panel **18** is disposed in a position offset from the optical axis N of the first optical system **31** toward the upper side Y**1**. The projection image is therefore formed in a position offset from the optical axis N toward the upper side Y**1**.
(81) The beams from the liquid crystal panel **18** pass through the first optical system **31** and the

second optical system **32** in the presented order. Between the first optical system **31** and the second optical system **32**, the beams pass through the lower side Y**2** of the optical axis N. The beams are therefore directed through the second optical system **32** toward the reflection surface **40**. The beams having reached the reflection surface **40** are deflected back in the first direction Z**1** towards the upper side Y**1**. The beams deflected back by the reflection surface **40** cross the optical axis N toward the upper side Y**1** and travel toward the first lens **34**. The beams passing through the first lens **34** are widened by the first lens **34** and reach the screen S.

(82) The lens L**16** of the first optical system **31** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. An intermediate image **30** is formed between the lens L**16** and the reflection surface **40**.

(83) In the projection system **3**B, the portion at the reduction side of the first optical system **31** is a telecentric portion.

(84) The projection system **3**B has a changeable projection distance. When the projection distance is changed, seven lenses of the first optical system **31**, the lenses L**10** to L**16**, are moved along the optical axis N for focusing. In the focusing, the lenses L**12**, L**13**, and L**14** are moved as a unit.

(85) Data on the projection system **3**B are listed below,

(86) TABLE-US-00007 NA 0.2778 imy 11.8 mm scy 1462 mm PD 288.6 mm M 124 TR 0.197 OAL 189 mm LL 36.5 mm

where NA represents the numerical aperture of the liquid crystal panel **18**, imy represents a first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, scy represents a second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, PD represents a projection distance that is the distance from the first lens **34** to the screen S, M represents a projection magnification that is the quotient of division of the second distance by the first distance, TR represents a throw ratio that is the quotient of division of the projection distance by the second distance, OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, and LL represents the largest radius of the first lens **34**.

(87) Data on the lenses of the projection system **3**B are listed below. The surfaces of the lenses are numbered sequentially from the reduction side toward the enlargement side. Reference characters are given to the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen. Data labeled with a surface number that does not correspond to any of the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen is dummy data. Reference character R represents the radius of curvature. Reference character D represents the axial inter-surface spacing. Reference character C represents the aperture radius, and twice the aperture radius is the diameter of the lens surface. Reference characters R, D, and C are each expressed in millimeters.

(88) TABLE-US-00008 Reference Surface Refraction/ character number Shape R D Glass material Reflection C 18 0 Spherical Infinity 9.5000 Refraction 0.0000 19 1 Spherical Infinity 25.9100 SBSL7_OHARA Refraction 12.8706 2 Spherical Infinity 0.0000 Refraction 14.7871 L1 3 Spherical 23.3077 7.9094 SFPL51_OHARA Refraction 15.4467 4 Spherical −127.0559 0.1000 Refraction 15.2079 L2 5 Spherical 21.0056 5.4283 SFPL51-OHARA Refraction 13.0000 L3 6 Spherical 44.2633 1.2142 STIH6_OHARA Refraction 11.8639 7 Spherical 26.7444 0.1000 Refraction 11.0399 L4 8 Spherical 17.2046 7.4036 SFPL51_OHARA Refraction 10.5313 L5 9 Spherical −23.7897 0.7500 TAFD25_HOYA Refraction 9.6431 10 Spherical 20.7471 0.2000 Refraction 8.5696 L6 11 Aspherical 14.2713 4.2460 LBAL35_OHARA Refraction 8.6504 12 Aspherical 32.7925 1.0000 Refraction 7.7784 L7 13 Spherical 16.4731 2.7568 SFSL5_OHARA Refraction 7.6827 14 Spherical 23.3002 2.7433 Refraction 7.2371 51 15 Spherical Infinity 0.1000 Refraction 6.9632 L8 16 Spherical 55.5146 3.5141 STIH53_OHARA Refraction 7.2124 17 Spherical −20.4634 0.1000 Refraction 7.4622 L9 18 Aspherical −18.8417 4.8490 LLAM60_OHARA Refraction 7.4571 19 Aspherical 138.4003 Variable Refraction 8.3506 spacing

1 L10 20 Spherical 77.9934 3.2176 STIL25_OHARA Refraction 13.5492 L11 21 Spherical −238.3916 1.0000 STIH6_OHARA Refraction 13.7984 22 Spherical 422.0121 Variable Refraction 14.0280 spacing 2 L12 23 Spherical 35.5651 10.2093 STIM22_OHARA Refraction 15.5000 24 Spherical −34.9646 0.7658 Refraction 15.5950 L13 25 Spherical −38.6490 5.7518 STIL25_OHARA Refraction 15.1870 L14 26 Spherical −19.2778 1.0000 STIH6_OHARA Refraction 15.1633 27 Spherical 431.4628 Variable Refraction 16.3418 spacing 3 L15 28 Aspherical −22.5593 3.0000 ‘Z-E48R’ Refraction 16.5057 29 Aspherical 111.9646 Variable Refraction 18.7911 spacing 4 L16 30 Aspherical 245.8978 8.0000 ‘Z-E48R’ Refraction 24.6934 31 Aspherical 58.0779 Variable Refraction spacing 5 40 32 Aspherical −28.3812 −46.6942 Reflection 39.1428 34 33 Aspherical 33.3357 −7.0000 ‘Z-E48R’ Refraction 27.0468 34 Aspherical 30.6616 Variable Refraction 36.2389 spacing 6 S 35 Spherical Infinity 0.0000 Refraction 2001.4151

(89) The projection system **3**B according to the present example has a changeable projection distance selected from a standard distance, a short distance shorter than the standard distance, and a long distance longer than the standard distance. When the projection distance is changed, seven lenses of the first optical system **31**, the lenses L**10** to L**16**, are moved along the optical axis N for focusing. When the focusing is performed so as to change the projection distance from the short distance to the long distance, the lenses L**10** and L**11** move along the optical axis N toward the reduction side. In the same focusing operation, the lenses L**12**, L**13**, and L**14** move along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**15** moves along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**16** moves along the optical axis N toward the enlargement side.

(90) The table below shows the variable spacings 1, 2, 3, 4, 5, and 6 at the projection distances where the focusing is performed. The variable spacing 1 is the axial inter-surface spacing between the lens L**9** and the lens L**10**. The variable spacing 2 is the axial inter-surface spacing between the lens L**11** and the lens L**12**. The variable spacing 3 is the axial inter-surface spacing between the lens L**14** and the lens L**15**. The variable spacing 4 is the axial inter-surface spacing between the lens L**15** and the lens L**16**. The variable spacing 5 is the axial inter-surface spacing between the lens L**16** and the reflection surface **40**. The variable spacing 6 is the projection distance.

(91) TABLE-US-00009 Standard Short Long distance distance distance Variable spacing 1 12.1167 13.2136 11.8202 Variable spacing 2 1.8056 0.1000 2.7521 Variable spacing 3 5.1195 5.3664 4.9687 Variable spacing 4 20.8516 21.2210 20.3846 Variable spacing 5 38.8253 38.8412 38.8165 Variable spacing 6 −287.0000 −224.0000 −403.0000

(92) The aspherical coefficients are listed below.

(93) TABLE-US-00010 Surface number S11 S12 S18 S19 Radius of 14.2713 32.7925 −18.8417 138.4003 curvature (R) Conic constant (K) 8.61373E−01 1.26960E+01 −1 −90 Fourth-order −1.78411E−05 1.08502E−04 6.36736E−05 9.09981E−05 Sixth-order 1.92270E−08 3.62039E−07 −9.07712E−07 −7.42228E−07 Eighth-order 8.84500E−11 −2.22710E−09 3.81595E−09 3.02545E−09 Tenth-order 4.90294E−11 Surface number S28 S29 S30 S31 Radius of −22.5593 111.9646 245.8978 58.0779 curvature (R) Conic constant (K) 0 0 90 0.00000E+00 Fourth-order 1.20307E−04 1.83761E−05 −5.27611E−05 −8.10795E−05 Sixth-order −6.19796E−07 −2.79021E−07 1.51125E−07 2.09269E−07 Eighth-order 1.66495E−09 6.93645E−10 −3.07259E−10 −3.89250E−10 Tenth-order −1.73986E−12 −6.65573E−13 4.39273E−13 4.46405E−13 Twelfth-order −2.61547E−16 −1.98979E−16 Fourteenth-order 6.56482E−22 Surface number S32 S33 S34 Radius of −28.3812 33.3357 30.66161366 curvature (R) Conic constant (K) −1.00000E+00 −5.39402E−01 −0.60807081 Fourth-order 3.57234E−06 1.50335E−04 4.29399E−05 Sixth-order −6.79154E−09 −9.61493E−07 −2.02768E−07 Eighth-order 9.60704E−12 3.12494E−09 3.95931E−10 Tenth-order −7.92734E−15 −5.89295E−12 −4.31475E−13 Twelfth-order 3.36716E−18 6.57780E−15 2.74639E−16 Fourteenth-order −6.07598E−22 −4.02279E−18

−9.56386E−20 Sixteenth-order 1.04005E−21 1.41923E−23

(94) The projection system **3**B according to the present example satisfies all Conditional Expressions (1) and (2) below,

$TR \leq 0.3$ (1)

$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60$ (2)

where OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, imy represents the first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, LL represents the largest radius of the first lens **34**, TR represents the throw ratio, which is the quotient of division of the projection distance by the second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, and NA represents the numerical aperture of the liquid crystal panel **18**.

(95) Furthermore, it is more preferable that the projection system **3**B satisfies all Conditional Expressions (1) and (2′) below.

$TR \leq 0.3$ (1)

$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 53$ (2′)

(96) In the present example, the values described above are listed below.

(97) TABLE-US-00011 OAL 189 mm imy 11.8 mm LL 36.5 mm TR 0.197 NA 0.2778

(98) TR=0.197 is provided from the table shown above, so that Conditional Expression (1) is satisfied. (OAL/imy)×(LL/imy)×TR×(1/NA)=35 is satisfied, so that Conditional Expression (2) is satisfied.

Effects and Advantages

(99) In the projection system **3**B according to the present example, the reflection surface **40** is provided with a reflection coating layer (reflection layer). The projection system **3**B according to the present example can therefore provide the same effects and advantages as those provided by Example 1.

(100) In the projection system **3**B according to the present example, the lens L**16** (second lens) disposed at a position closest to the enlargement side in the first optical system **31** is formed as a lens separate from the first lens **34**. The lens L**16** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. That is, the lens L**16** disposed at a position closest to the enlargement side in the first optical system **31** is disposed inside the second optical system **32** in the direction of the optical axis N, so that the distance between the lens L**16** and the reflection surface **40** decreases. The projection system **3**B according to the present example can therefore provide the same effects and advantages as those provided by Example 1.

(101) The first optical system **31** includes the lens L**16** (second lens) and the lens L**15** (third lens), which is disposed adjacent to the lens L**16** and shifted therefrom toward the reduction side. The lenses L**15** and L**16** each have an aspherical shape. In the projection system **3**B according to the present example, focusing that causes the projection distance to be changed from the short distance to the long distance is performed by moving the lenses L**15** and L**16** toward the enlargement side in the direction of the optical axis N. The projection system **3**B, in which the lenses L**15** and L**16**, which correct a variety of aberrations on an image height basis, are moved in the direction of the optical axis N, therefore allows suppression of occurrence of the variety of aberrations during focusing. Furthermore, in a configuration in which a lens having no aspherical shape is moved in the direction of the optical axis N for focusing, an aspherical lens that corrects a variety of aberrations needs to be separately prepared. In contrast, the present example, in which the lenses L**15** and L**16**, which move during focusing, each have an aspherical shape, allows reduction in the size of the entire projection system.

(102) The first optical system **31** further includes the cemented doublets L**23** and L**24** at the enlargement side of the diaphragm **51**. The chromatic aberrations can therefore be corrected well.

(103) The projection system **3**B according to the present example, which satisfies Conditional Expressions (1) and (2), can provide the same effects and advantages as those provided by the

projection system **3**A according to Example 1. FIG. **8** shows the lateral aberrations produced by the projection system **3**B set at the standard distance. FIG. **9** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**B set at the standard distance. FIG. **10** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**B set at the short distance. FIG. **11** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**B set at the long distance. The projection system **3**B according to the present example produces an enlarged image having suppressed aberrations, as shown in FIGS. **8** to **11**.

Example 3

(104) FIG. **12** is a beam diagram showing beams passing through a projection system **3**C according to Example 3. The projection system **3**C according to the present example is formed of a first optical system **31** and a second optical system **32** sequentially arranged from the reduction side toward the enlargement side, as shown in FIG. **12**. The second optical system **32** is disposed on an optical axis N of the first optical system **31**.

(105) The first optical system **31** is a refractive optical system. The first optical system **31** is formed of thirteen lenses L**1** to L**13**. The lenses L**1** to L**13** are arranged in the presented order from the reduction side toward the enlargement side. A diaphragm **51** is disposed between the lens L**6** and the lens L**7**.

(106) The lens L**5** has aspherical shapes at opposite sides. The lens L**8** has aspherical shapes at opposite sides. The lens L**12** (third lens) has aspherical shapes at opposite sides. The lens L**13** (second lens) has aspherical shapes at opposite sides. The lens L**3** and the lens L**4** are bonded to each other into a cemented doublet L**21**. The lens L**10** and the lens L**11** are bonded to each other into a cemented doublet L**22**.

(107) The second optical system **32** includes an optical element **33** and a first lens **34**. The optical element **33** and the first lens **34** are arranged in the presented order from the reduction side toward the enlargement side. The optical element **33** has a reflection surface **40**, which faces the reduction side. The reflection surface **40** has a concave shape recessed in the second direction Z**2**. The reflection surface **40** has an aspherical shape. The reflection surface **40** is located at the lower side Y**2** of the optical axis N, as shown in FIG. **12**. The reflection surface **40** is formed by providing the outer surface, in the first direction Z**1**, of the optical element **33** with a reflection coating layer (reflection layer). The reflection surface **40** reflects light at the surface, facing in the direction Z**1**, of the optical element **33**.

(108) The first lens **34** is located at a position shifted from the optical element **33** in the first direction Z**1** and disposed at the upper side Y**1** of the optical axis N. The first lens **34** has negative power. The first lens **34** has a convex enlargement-side surface and a concave reduction-side surface. The first lens **34** has aspherical shapes at opposite sides.

(109) The liquid crystal panel **18** of the image formation unit **2** is disposed in the reduction-side conjugate plane of the projection system **3**C. The screen S is disposed in the enlargement-side conjugate plane of the projection system **3**C.

(110) The liquid crystal panel **18** forms a projection image in an image formation plane perpendicular to the optical axis N of the first optical system **31**. The liquid crystal panel **18** is disposed in a position offset from the optical axis N of the first optical system **31** toward the upper side Y**1**. The projection image is therefore formed in a position offset from the optical axis N toward the upper side Y**1**.

(111) The beams from the liquid crystal panel **18** pass through the first optical system **31** and the second optical system **32** in the presented order. Between the first optical system **31** and the second optical system **32**, the beams pass through the lower side Y**2** of the optical axis N. The beams are therefore directed through the second optical system **32** toward the reflection surface **40**. The beams having reached the reflection surface **40** are deflected back in the first direction Z**1** towards the upper side Y**1**. The beams deflected back by the reflection surface **40** cross the optical axis N

toward the upper side Y**1** and travel toward the first lens **34**. The beams passing through the first lens **34** are widened by the first lens **34** and reach the screen S.
(112) The lens L**13** of the first optical system **31** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. An intermediate image **30** is formed between the lens L**13** and the reflection surface **40**.
(113) In the projection system **3**C, the portion at the reduction side of the first optical system **31** is a telecentric portion.
(114) The projection system **3**C has a changeable projection distance. When the projection distance is changed, four lenses of the first optical system **31**, the lenses L**9** to L**12**, are moved along the optical axis N for focusing. In the focusing, the lenses L**10** and L**11** are moved as a unit.
(115) Data on the projection system **3**C are listed below,
(116) TABLE-US-00012 NA 0.2084 imy 11.7 mm scy 1462 mm PD 378.0 mm M 125 TR 0.259 OAL 172 mm LL 37.8 mm
where NA represents the numerical aperture of the liquid crystal panel **18**, imy represents a first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, scy represents a second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, PD represents a projection distance that is the distance from the first lens **34** to the screen S, M represents a projection magnification that is the quotient of division of the second distance by the first distance, TR represents a throw ratio that is the quotient of division of the projection distance by the second distance, OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, and LL represents the largest radius of the first lens **34**.
(117) Data on the lenses of the projection system **3**C are listed below. The surfaces of the lenses are numbered sequentially from the reduction side toward the enlargement side. Reference characters are given to the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen. Data labeled with a surface number that does not correspond to any of the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen is dummy data. Reference character R represents the radius of curvature. Reference character D represents the axial inter-surface spacing. Reference character C represents the aperture radius, and twice the aperture radius is the diameter of the lens surface. Reference characters R, D, and C are each expressed in millimeters.
(118) TABLE-US-00013 Reference Surface Glass Refraction/ character number Shape R D material Reflection C 18 0 Spherical Infinity 5.1000 Refraction 0.0000 19 1 Spherical Infinity 23.9250 SBSL7_OHARA Refraction 12.0558 2 Spherical Infinity 0.0000 Refraction 13.1538 L1 3 Spherical 25.3616 5.2165 SFPL51_OHARA Refraction 13.4219 4 Spherical −160.4483 0.1000 Refraction 13.2933 L2 5 Spherical 20.5113 4.3359 SFSL5_OHARA Refraction 12.0000 6 Spherical 132.0437 0.0000 Refraction 11.5962 7 Spherical Infinity 0.1000 Refraction 11.9120 L3 8 Spherical 20.3053 7.3344 SBSL7_OHARA Refraction 10.1626 L4 9 Spherical −23.1488 0.6000 TAFD25_HOYA Refraction 8.3549 10 Spherical 13.5441 0.2000 Refraction 7.2414 L5 11 Aspherical 10.7115 2.4947 LBAL35_OHARA Refraction 7.2839 12 Aspherical 21.0609 8.3786 Refraction 6.9453 L6 13 Spherical 34.9176 3.6553 SFSL5_OHARA Refraction 6.3040 14 Spherical −16.1488 0.0364 Refraction 6.0810 51 15 Spherical Infinity 0.1000 Refraction 5.5409 L7 16 Spherical −371.7544 2.2081 STIH53_OHARA Refraction 5.5362 17 Spherical −27.0329 0.1512 Refraction 5.4751 L8 18 Aspherical −23.1991 2.3150 LLAM60_OHARA Refraction 5.4355 19 Aspherical 89.7692 Variable Refraction 5.5698 spacing 1 L9 20 Spherical −64.9866 5.7941 STIM22_OHARA Refraction 15.7747 21 Spherical −27.0076 Variable Refraction 16.5319 spacing 2 L10 22 Spherical 43.6185 10.6351 STIL25_OHARA Refraction 17.8154 L11 23 Spherical −30.6037 1.0000 STIH6_OHARA Refraction 17.7153 24 Spherical 66.9444 Variable Refraction 17.8736 spacing 3 L12 25 Aspherical −49.3948 6.5927 ‘Z-E48R’ Refraction 19.0969 26 Aspherical 33.1964 Variable Refraction 18.5842 spacing 4 L13 27 Aspherical 198.8399 5.0000 ‘Z-E48R’

Refraction 18.7753 28 Aspherical 41.3371 35.9841 Refraction 19.8477 40 29 Aspherical −28.5883 −41.4150 Reflection 29.2645 34 30 Aspherical 38.8043 −5.0000 ‘Z-E48R’ Refraction 27.5293 31 Aspherical 39.6045 Variable Refraction 36.7982 spacing 5 S 32 Spherical Infinity 0.0000 Refraction 1983.0341

(119) The projection system **3**C according to the present example has a changeable projection distance selected from a standard distance, a short distance shorter than the standard distance, and a long distance longer than the standard distance. When the projection distance is changed, four lenses of the first optical system **31**, the lenses L**9** to L**12**, are moved along the optical axis N for focusing. When the focusing is performed so as to change the projection distance from the short distance to the long distance, the lens L**9** moves along the optical axis N toward the enlargement side. In the same focusing operation, the lenses L**10** and L**11** move along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**12** moves along the optical axis N toward the enlargement side. In the projection system **3**C according to the present example, the lens L**13** is fixed.

(120) The table below shows the variable spacings 1, 2, 3, 4, and 5 at the projection distances where the focusing is performed. The variable spacing 1 is the axial inter-surface spacing between the lens L**8** and the lens L**9**. The variable spacing 2 is the axial inter-surface spacing between the lens L**9** and the lens L**10**. The variable spacing 3 is the axial inter-surface spacing between the lens L**11** and the lens L**12**. The variable spacing 4 is the axial inter-surface spacing between the lens L**12** and the lens L**13**. The variable spacing 5 is the projection distance.

(121) TABLE-US-00014 Standard Short Long distance distance distance Variable spacing 1 27.7058 27.3443 27.9475 Variable spacing 2 0.2085 0.0000 0.4035 Variable spacing 3 5.0259 5.2821 4.8227 Variable spacing 4 8.2762 8.5655 8.0181 Variable spacing 5 −379.0000 −297.0000 −524.0000

(122) The aspherical coefficients are listed below.

(123) TABLE-US-00015 Surfaces number S11 S12 S18 S19 Radius of 10.7115 21.0609 −23.1991 89.7692 curvature (R) Conic constant (K) 2.69624E−01 4.78408E+00 −1 −90 Fourth-order −7.67687E−05 5.05163E−05 1.47441E−04 2.11126E−04 Sixth-order −5.77986E−07 −3.02482E−07 −2.37490E−06 −2.24828E−06 Eighth-order −4.01080E−09 −4.23268E−09 1.45245E−08 1.46576E−08 Tenth-order 5.29887E−11 Surface number S25 S26 S27 S28 Radius of −49.3948 33.1964 198.8399 41.3371 curvature (R) Conic constant (K) 0 0 90 0.00000E+00 Fourth-order 6.80021E−05 −4.70087E−05 −6.65376E−05 −1.21300E−04 Sixth-order −9.65755E−08 1.44566E−07 −7.25627E−09 4.65229E−07 Eighth-order 7.76122E−11 −3.72543E−10 6.19846E−10 −1.09606E−09 Tenth-order 2.18093E−14 4.76925E−13 −1.08500E−12 1.26985E−12 Twelfth-order 3.76822E−16 2.28175E−16 Fourteenth-order −1.41383E−18 Surface number S29 S30 S31 Radius of −28.5883 38.8043 39.60445722 curvature (R) Conic constant (K) −1.00000E+00 4.72790E−01 −0.611371609 Fourth-order 3.91548E−06 −1.70359E−05 −2.40495E−05 Sixth-order −1.20322E−08 8.73558E−08 5.12452E−08 Eighth-order 2.02386E−11 −2.61450E−10 −6.11798E−11 Tenth-order −2.19615E−14 5.37788E−13 4.72800E−14 Twelfth-order 1.28670E−17 −6.50975E−16 −2.21909E−17 Fourteenth-orde −3.20304E−21 4.05800E−19 5.45183E−21 Sixteenth-order −9.14385E−23 −3.94213E−25

(124) The projection system **3**C according to the present example satisfies all Conditional Expressions (1) and (2) below,

$TR \leq 0.3$ (1)

$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60$ (2)

where OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, imy represents the first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, LL represents the largest radius of the first lens **34**, TR represents the throw ratio, which is the quotient of division of the projection distance by the second

distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, and NA represents the numerical aperture of the liquid crystal panel **18**.
(125) In the present example, the values described above are listed below.
(126) TABLE-US-00016 OAL 172 mm imy 11.7 mm LL 37.8 mm TR 0.259 NA 0.2084
(127) TR=0.259 is provided from the table shown above, so that Conditional Expression (1) is satisfied. (OAL/imy)×(LL/imy)×TR×(1/NA)=59 is satisfied, so that Conditional Expression (2) is satisfied.
Effects and Advantages
(128) In the projection system **3**C according to the present example, the reflection surface **40** is provided with a reflection coating layer (reflection layer). The projection system **3**C according to the present example can therefore provide the same effects and advantages as those provided by Example 1.
(129) In the projection system **3**C according to the present example, the lens L**13** (second lens) disposed at a position closest to the enlargement side in the first optical system **31** is formed as a lens separate from the first lens **34**. The lens L**13** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. That is, the lens L**13** disposed at a position closest to the enlargement side in the first optical system **31** is disposed inside the second optical system **32** in the direction of the optical axis N, so that the distance between the lens L**13** and the reflection surface **40** decreases. The projection system **3**C according to the present example can therefore provide the same effects and advantages as those provided by Example 1.
(130) In the projection system **3**C according to the present example, the first optical system **31** includes the lens L**13** (second lens) and the lens L**12** (third lens), which is disposed adjacent to the lens L**13** and shifted therefrom toward the reduction side. That is, the lens L**13** disposed at a position closest to the enlargement side in the first optical system **31** is disposed inside the second optical system **32** in the direction of the optical axis N, so that the distance between the lens L**13** and the reflection surface **40** decreases. In the projection system **3**C according to the present example, focusing that causes the projection distance to be changed from the short distance to the long distance is performed by moving the lens L**12** toward the enlargement side in the direction of the optical axis N. The projection system **3**C, in which the lens L**12**, which corrects a variety of aberrations on an image height basis, is moved in the direction of the optical axis N, therefore allows suppression of occurrence of the variety of aberrations during focusing. Furthermore, in a configuration in which a lens having no aspherical shape is moved in the direction of the optical axis N for focusing, an aspherical lens that corrects a variety of aberrations needs to be separately prepared. In contrast, the present example, in which the lens L**12**, which moves during focusing, has an aspherical shape, allows reduction in the size of the entire projection system.
(131) The lens L**13** is fixed in the direction of the optical axis N. The lens L**13**, which is the second lens, is disposed between the reflection surface **40** and the first lens **34**. Therefore, when the lens L**13** is moved in the direction of the optical axis N for focusing, the mechanism that moves the lens L**13** is complicated, resulting in an increase in manufacturing cost. The manufacturing cost of the projection system **3**C according to the present example can therefore be suppressed as compared with the manufacturing cost of the projection system **3**A according to Example 1, in which the lens L**17**, which is the second lens, is moved in the direction of the optical axis N.
(132) The first optical system **31** further includes the cemented doublet L**22** at the enlargement side of the diaphragm **51**. The chromatic aberrations can therefore be corrected well.
(133) The projection system **3**C according to the present example, which satisfies Conditional Expressions (1) and (2), can provide the same effects and advantages as those provided by the projection system **3**A according to Example 1. FIG. **13** shows the lateral aberrations produced by the projection system **3**C set at the standard distance. FIG. **14** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**C set at the standard distance. FIG. **15** shows the spherical aberration, astigmatism, and distortion produced by the projection system

3C set at the short distance. FIG. **16** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**C set at the long distance. The projection system **3**C according to the present example produces an enlarged image having suppressed aberrations, as shown in FIGS. **13** to **16**.

Example 4

(134) FIG. **17** is a beam diagram showing beams passing through a projection system **3**D according to Example 4. The projection system **3**D according to the present example is formed of a first optical system **31** and a second optical system **32** sequentially arranged from the reduction side toward the enlargement side, as shown in FIG. **17**. The second optical system **32** is disposed on an optical axis N of the first optical system **31**.

(135) The first optical system **31** is a refractive optical system. The first optical system **31** is formed of thirteen lenses L**1** to L**13**. The lenses L**1** to L**13** are arranged in the presented order from the reduction side toward the enlargement side. A diaphragm **51** is disposed between the lens L**6** and the lens L**7**.

(136) The lens L**5** has aspherical shapes at opposite sides. The lens L**8** has aspherical shapes at opposite sides. The lens L**12** (third lens) has aspherical shapes at opposite sides. The lens L**13** (second lens) has aspherical shapes at opposite sides. The lens L**3** and the lens L**4** are bonded to each other into a cemented doublet L**21**. The lens L**10** and the lens L**11** are bonded to each other into a cemented doublet L**22**.

(137) The second optical system **32** includes an optical element **33** and a first lens **34**. The optical element **33** and the first lens **34** are arranged in the presented order from the reduction side toward the enlargement side. The optical element **33** has a first surface **36**, which faces the reduction side, and a second surface **37**, which faces the side opposite from the first surface **36**. The optical element **33** has a reflection coating layer at the second surface **37**. The first surface **36** has a concave shape. The second surface **37** has a convex shape. The optical element **33** has a first transmission surface **41**, a reflection surface **42**, and a second transmission surface **43** sequentially arranged from the reduction side toward the enlargement side. The first transmission surface **41** is provided at the first surface **36**. The first transmission surface **41** has a concave shape. The reflection surface **42** is the reflection coating layer and has a concave shape to which the surface shape of the second surface **37** has been transferred. The reflection surface **42** reflects light within the optical element **33**. The second transmission surface **43** is provided at the first surface **36**. The second transmission surface **43** has a concave shape. The first transmission surface **41**, the reflection surface **42**, and the second transmission surface **43** each have an aspherical shape. The first transmission surface **41**, the reflection surface **42**, and the second transmission surface **43** are located at the lower side Y**2** of the optical axis N, as shown in FIG. **17**.

(138) The first lens **34** is located at a position shifted from the optical element **33** in the first direction Z**1** and disposed at the upper side Y**1** of the optical axis N. The first lens **34** has negative power. The first lens **34** has a convex enlargement-side surface and a concave reduction-side surface. The first lens **34** has aspherical shapes at opposite sides.

(139) The liquid crystal panel **18** of the image formation unit **2** is disposed in the reduction-side conjugate plane of the projection system **3**D. The screen S is disposed in the enlargement-side conjugate plane of the projection system **3**D.

(140) The liquid crystal panel **18** forms a projection image in an image formation plane perpendicular to the optical axis N of the first optical system **31**. The liquid crystal panel **18** is disposed in a position offset from the optical axis N of the first optical system **31** toward the upper side Y**1**. The projection image is therefore formed in a position offset from the optical axis N toward the upper side Y**1**.

(141) The beams from the liquid crystal panel **18** pass through the first optical system **31** and the second optical system **32** in the presented order. Between the first optical system **31** and the second optical system **32**, the beams pass through the lower side Y**2** of the optical axis N. The beams are

thus incident on the first transmission surface **41** of the optical element **33**, which forms the second optical system **32**.

(142) The beams having entered the optical element **33** via the first transmission surface **41** travel toward the reflection surface **42**. The beams having reached the reflection surface **42** are deflected back in the first direction Z**1** towards the upper side Y**1**. The beams deflected back by the reflection surface **42** travel toward the second transmission surface **43**. The beams having exited via the second transmission surface **43** cross the optical axis N toward the upper side Y**1** and travel toward the first lens **34**. The beams passing through the first lens **34** are widened by the first lens **34** and reach the screen S.

(143) The lens L**13** of the first optical system **31** is disposed between the reflection surface **42** and the first lens **34** in the direction of the optical axis N. An intermediate image **30** is formed between the lens L**13** and the reflection surface **42**.

(144) In the projection system **3**D, the portion at the reduction side of the first optical system **31** is a telecentric portion.

(145) The projection system **3**D has a changeable projection distance. When the projection distance is changed, four lenses of the first optical system **31**, the lenses L**9** to L**12**, are moved along the optical axis N for focusing.

(146) Data on the projection system **3**D are listed below,

(147) TABLE-US-00017 NA 0.25 imy 11.7 mm scy 1463 mm PD 376.0 mm M 125 TR 0.257 OAL 175 mm LL 40.0 mm

where NA represents the numerical aperture of the liquid crystal panel **18**, imy represents a first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, scy represents a second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, PD represents a projection distance that is the distance from the first lens **34** to the screen S, M represents a projection magnification that is the quotient of division of the second distance by the first distance, TR represents a throw ratio that is the quotient of division of the projection distance by the second distance, OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **42**, and LL represents the largest radius of the first lens **34**.

(148) Data on the lenses of the projection system **3**D are listed below. The surfaces of the lenses are numbered sequentially from the reduction side toward the enlargement side. Reference characters are given to the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen. Data labeled with a surface number that does not correspond to any of the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen is dummy data. Reference character R represents the radius of curvature. Reference character D represents the axial inter-surface spacing. Reference character C represents the aperture radius, and twice the aperture radius is the diameter of the lens surface. Reference characters R, D, and C are each expressed in millimeters.

(149) TABLE-US-00018 Reference Surface Glass Refraction/ character number Shape R D material Reflection C 18 0 Spherical Infinity 5.1000 Refraction 0.0000 19 1 Spherical Infinity 23.9250 SBSL7_OHARA Refraction 12.2082 2 Spherical Infinity 0.0000 Refraction 13.7743 L1 3 Spherical 20.7308 7.6535 SFPL51_OHARA Refraction 14.3491 4 Spherical −160.5725 0.1403 Refraction 14.0489 L2 5 Spherical 19.8009 3.4530 SFSL5_OHARA Refraction 12.0000 6 Spherical 42.1883 0.0000 Refraction 11.4760 7 Spherical Infinity 0.1846 Refraction 12.5433 L3 8 Spherical 26.7793 5.8661 SBSL7_OHARA Refraction 10.8722 L4 9 Spherical −23.4816 0.7000 TAFD25_HOYA Refraction 10.1298 10 Spherical 23.8348 0.2000 Refraction 8.9729 L5 11 Aspherical 14.1866 3.5452 LBAL35_OHARA Refraction 8.8797 12 Aspherical 23.2921 8.5134 Refraction 8.1796 L6 13 Spherical 33.2276 4.4787 SFSL5_OHARA Refraction 7.8479 14 Spherical −16.5041 0.0000 Refraction 7.6677 51 15 Spherical Infinity 0.1000 Refraction 6.6414 L7 16 Spherical 846.5448 2.2054 STIH53_OHARA Refraction 6.6229 17 Spherical −40.4470 0.2313

Refraction 6.4863 L8 18 Aspherical −25.5044 3.6042 LLAM60_OHARA Refraction 6.4792 19 Aspherical 59.3482 Variable Refraction 6.1272 spacing 1 L9 20 Spherical −437.3885 6.0000 STIM22_OHARA Refraction 17.0050 21 Spherical −37.6532 Variable Refraction 17.5773 spacing 2 L10 22 Spherical 31.8187 11.5532 STIL25_OHARA Refraction 19.3821 L11 23 Spherical −47.9275 1.3832 STIH6_OHARA Refraction 19.2050 24 Spherical 49.3856 Variable Refraction 18.3019 spacing 3 L12 25 Aspherical −177.2317 2.0000 'Z-E48R' Refraction 18.4483 26 Aspherical 22.7112 Variable Refraction 17.9293 spacing 4 L13 27 Aspherical 179.6432 5.0000 'Z-E48R' Refraction 19.9221 28 Aspherical 27.1594 35.4677 Refraction 19.7715 41 29 Aspherical −32.3930 3.0000 'Z-E48R' Refraction 30.0301 42 30 Aspherical −30.4878 −3.0000 'Z-E48R' Reflection 30.7470 43 31 Aspherical −32.3930 −41.1512 Refraction 29.6535 34 32 Aspherical 39.4467 −5.0000 'Z-E48R' Refraction 35.7473 33 Aspherical 38.8749 Variable Refraction 39.9951 spacing 5 S 34 Spherical Infinity 0.0000 Refraction 1984.0193

(150) The projection system **3**D according to the present example has a changeable projection distance selected from a standard distance, a short distance shorter than the standard distance, and a long distance longer than the standard distance. When the projection distance is changed, four lenses of the first optical system **31**, the lenses L**9** to L**12**, are moved along the optical axis N for focusing. When the focusing is performed so as to change the projection distance from the short distance to the long distance, the lens L**9** moves along the optical axis N toward the reduction side. In the same focusing operation, the lenses L**10** and L**11** move along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**12** moves along the optical axis N toward the enlargement side. In the projection system **3**D according to the present example, the lens L**13** is fixed.

(151) The table below shows the variable spacings 1, 2, 3, 4, and 5 at the projection distances where the focusing is performed. The variable spacing 1 is the axial inter-surface spacing between the lens L**8** and the lens L**9**. The variable spacing 2 is the axial inter-surface spacing between the lens L**9** and the lens L**10**. The variable spacing 3 is the axial inter-surface spacing between the lens L**11** and the lens L**12**. The variable spacing 4 is the axial inter-surface spacing between the lens L**12** and the lens L**13**. The variable spacing 5 is the projection distance.

(152) TABLE-US-00019

	Standard distance	Short distance	Long distance
Variable spacing 1	26.8792	27.0101	26.6608
Variable spacing 2	1.4097	0.7506	2.2492
Variable spacing 3	2.1500	2.6303	1.5624
Variable spacing 4	10.4814	10.5279	10.4465
Variable spacing 5	−376.0000	−295.0000	−519.0000

(153) The aspherical coefficients are listed below.

(154) TABLE-US-00020

Surface number	S11	S12	S18	S19
Radius of curvature (R)	14.1866	23.2921	25.5044	59.3482
Conic constant (K)	0.52162588	4.313809097	−1	−90
Fourth-order	−2.90353E−05	1.01864E−04	9.38265E−05	1.86743E−04
Sixth-order	−2.33563E−07	−1.24711E−06	−3.97479E−08	−1.68413E−06
Eighth-order	−7.08031E−09	5.72359E−09	9.15280E−09	−1.14146E−08
Tenth-order	2.69207E−11			

Surface number	S25	S26	S27	S28
Radius of curvature (R)	−177.2317	22.7112	179.6432	27.1594
Conic constant (K)	0	0	89.447202	0.000000
Fourth-order	7.96862E−05	−2.58626E−08	−6.20694E−05	−1.51197E−04
Sixth-order	−3.07379E−07	−1.92895E−07	4.44832E−07	9.77046E−07
Eighth-order	6.58543E−10	2.75182E−10	−2.36330E−09	−4.47460E−09
Tenth-order	−4.83689E−13	−1.73130E−15	5.49541E−12	1.16337E−11
Twelfth-order	−4.88375E−15	−1.62518E−14		
Fourteenth-order	9.38335E−18			

Surface number	S29, S31	S30	S32	S33
Radius of curvature (R)	−32.3930	−30.4878	39.4467	38.8749
Conic constant (K)	−1.026766	−1.000000	−0.081546	−0.618360
Fourth-order	9.61441E−06	5.12974E−06	−1.83922E−05	−1.83960E−05
Sixth-order	−4.77776E−08	−1.81978E−08	8.40433E−08	3.72462E−08
Eighth-order	1.03751E−10	2.96797E−11	−1.91747E−10	−5.13686E−11
Tenth-order	−1.17673E−13	−2.63402E−14	2.17884E−13	5.06109E−14
Twelfth-order	6.81981E−17	1.12362E−17	−5.07382E−17	−3.06209E−17
Fourteenth-order	−1.55320E−20	−1.48819E−21	−7.93912E−20	

1.04462E−20 Sixteenth-order 4.02973E−23 −1.62167E−24
(155) The projection system **3**D according to the present example satisfies all Conditional Expressions (1) and (2) below,

TR≤0.3 (1)

35≤(OAL/imy)×(LL/imy)×TR×(1/NA)≤60 (2)

where OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **42**, imy represents the first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, LL represents the largest radius of the first lens **34**, TR represents the throw ratio, which is the quotient of division of the projection distance by the second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, and NA represents the numerical aperture of the liquid crystal panel **18**.

(156) Furthermore, it is more preferable that the projection system **3**D satisfies all Conditional Expressions (1) and (2′) below.

TR≤0.3 (1)

35≤(OAL/imy)×(LL/imy)×TR×(1/NA)≤53 (2′)

(157) In the present example, the values described above are listed below.

(158) TABLE-US-00021 OAL 175 mm imy 11.7 mm LL 40.0 mm TR 0.257 NA 0.25

TR=0.257 is provided from the table shown above, so that Conditional Expression (1) is satisfied.

(OAL/imy)×(LL/imy)×TR×(1/NA)=53 is satisfied, so that Conditional Expression (2) is satisfied.

Effects and Advantages

(159) In the projection system **3**D according to the present example, the lens L**13** (second lens) disposed at a position closest to the enlargement side in the first optical system **31** is formed as a lens separate from the first lens **34**. The lens L**13** is disposed between the reflection surface **42** and the first lens **34** in the direction of the optical axis N. That is, the lens L**13** disposed at a position closest to the enlargement side in the first optical system **31** is disposed inside the second optical system **32** in the direction of the optical axis N, so that the distance between the lens L**13** and the reflection surface **42** decreases. The projection system **3**D according to the present example can therefore provide the same effects and advantages as those provided by Example 1.

(160) In the projection system **3**D according to the present example, the first optical system **31** includes the lens L**13** (second lens) and the lens L**12** (third lens), which is disposed adjacent to the lens L**13** and shifted therefrom toward the reduction side. The lenses L**12** and L**13** each have an aspherical shape. In the projection system **3**D according to the present example, focusing that causes the projection distance to be changed from the short distance to the long distance is performed by moving the lens L**12** toward the enlargement side in the direction of the optical axis N. The projection system **3**D, in which the lens L**12**, which corrects a variety of aberrations on an image height basis, is moved in the direction of the optical axis N, therefore allows suppression of occurrence of the variety of aberrations during focusing. Furthermore, in a configuration in which a lens having no aspherical shape is moved in the direction of the optical axis N for focusing, an aspherical lens that corrects a variety of aberrations needs to be separately prepared. In contrast, the present example, in which the lens L**12**, which moves during focusing, has an aspherical shape, allows reduction in the size of the entire projection system.

(161) The lens L**13** is fixed in the direction of the optical axis N. The lens L**13**, which is the second lens, is disposed between the reflection surface **42** and the first lens **34**. Therefore, when the lens L**13** is moved in the direction of the optical axis N for focusing, the mechanism that moves the lens L**13** is complicated, resulting in an increase in manufacturing cost. The manufacturing cost of the projection system **3**D according to the present example can therefore be suppressed as compared with the manufacturing cost of the projection system **3**A according to Example 1, in which the lens L**17**, which is the second lens, is moved in the direction of the optical axis N.

(162) The first optical system **31** further includes the cemented doublet L**22** at the enlargement side of the diaphragm **51**. The chromatic aberrations can therefore be corrected well.

(163) The projection system **3**D according to the present example, which satisfies Conditional Expressions (1) and (2), can provide the same effects and advantages as those provided by the projection system **3**A according to Example 1. FIG. **18** shows the lateral aberrations produced by the projection system **3**D set at the standard distance. FIG. **19** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**D set at the standard distance. FIG. **20** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**D set at the short distance. FIG. **21** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**D set at the long distance. The projection system **3**D according to the present example produces an enlarged image having suppressed aberrations, as shown in FIGS. **18** to **21**.

Claims

1. A projection system for enlarging a projection image formed by an image formation device disposed in a reduction-side conjugate plane and projecting the enlarged image in an enlargement-side conjugate plane, the projection system comprising: a first optical system and a second optical system sequentially arranged from the reduction side toward the enlargement side, wherein the first optical system includes a diaphragm, the second optical system includes an optical element having a concave reflection surface and a first lens having negative power, the optical element and the first lens sequentially arranged from the reduction side toward the enlargement side, an intermediate image conjugate with the reduction-side conjugate plane and the enlargement-side conjugate plane is formed between the first optical system and the second optical system, a portion at the reduction side of the first optical system forms a telecentric portion, and the projection system satisfies all Conditional Expressions (1) and (2) below,
$TR \leq 0.3$ (1)
$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60$ (2) where OAL represents an axial inter-surface spacing from the image formation device to the reflection surface, imy represents a first distance from an optical axis to a largest image height at the image formation device, LL represents a largest radius of the first lens, TR represents a throw ratio that is a quotient of division of a projection distance by a second distance from the optical axis to a largest image height of the enlarged image, and NA represents a numerical aperture of the image formation device.
2. The projection system according to claim 1, wherein the reflection surface is provided with a reflection layer.
3. The projection system according to claim 1, wherein a second lens disposed at a position closest to the enlargement side in the first optical system is formed as a lens separate from the first lens, and the second lens is disposed between the reflection surface and the first lens in a direction of the optical axis.
4. The projection system according to claim 3, wherein the first optical system includes the second lens and a third lens that is disposed adjacent to the second lens and shifted therefrom toward the reduction side, the second and third lenses each have an aspherical shape, and focusing that causes the projection distance to be changed from a short distance to a long distance is performed by moving the third lens toward the enlargement side in the direction of the optical axis.
5. The projection system according to claim 4, wherein the second lens is fixed in the direction of the optical axis.
6. The projection system according to claim 1, wherein the first optical system includes a cemented doublet on the enlargement side of the diaphragm.
7. A projector comprising: the projection system according to claim 1; and the image formation device that forms a projection image in the reduction-side conjugate plane of the projection system.